NATHAN [illegible]
(2231349089)
1C - TNK

No.

Date:

1. a) Assuming that water has a density of exactly $1\ g/cm^3$, find the mass of one cubic meter of water in kilograms.

$\Rightarrow \rho = \left(\frac{1\ g}{cm^3}\right)\left(\frac{1\ kg}{1000\ g}\right)\left(\frac{100\ cm}{1\ m}\right)^3$

$= 1000\ kg/m^3$

$\Rightarrow m = \rho . V$

$= 1000\ kg/m^3 . m^3$

$= 1000\ kg$

b) Suppose that it takes 10.0 h to drain a container of $5700\ m^3$ of water. What is the "mass flow rate" in kilograms per second, of water from the container?

$\Rightarrow m = \rho . V$

$= 1000\ kg/m^3 . 5700\ m^3$

$= 5,7 \times 10^6\ kg$

$\Rightarrow t = 10\ jam$

$= 10 . 3600 = 36.000\ sekon$

$\Rightarrow$ mass flow rate $= \frac{\text{massa}}{\text{waktu}} = \frac{5,7 \times 10^6}{36.000}$

$= 158,3\ kg/s$

2. Three vectors are given by $\vec{a} = 3\hat{i} + 3\hat{j} - 2\hat{k}$; $\vec{b} = -\hat{i} - 4\hat{j} + 2\hat{k}$, and $\vec{c} = 2\hat{i} + 2\hat{j} + 1\hat{k}$. Find (a) $\vec{a} . (\vec{b} \times \vec{c})$, (b) $\vec{a} . (\vec{b} + \vec{c})$, and (c) $\vec{a} \times (\vec{b} + \vec{c})$.

a) $\vec{a} . (\vec{b} \times \vec{c})$

$\Rightarrow \vec{b} \times \vec{c} =$

	$\hat{i}$	$\hat{j}$	$\hat{k}$
$\vec{b}$	-1	-4	2
$\vec{c}$	2	2	1

$= +\hat{i}\begin{vmatrix} -4 & 2 \\ 2 & 1 \end{vmatrix} - \hat{j}\begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} + \hat{k}\begin{vmatrix} -1 & -4 \\ 2 & 2 \end{vmatrix}$

$= +\hat{i}(-4.1 - (2.2)) - \hat{j}(-1.1 - (2.2)) + \hat{k}(-1.2 - (-4.2))$

$= -8\hat{i} + 5\hat{j} + 6\hat{k}$

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$\vec{a} \cdot (\vec{b} \times \vec{c}) \Rightarrow 3.(-8) + 3(5) - 2(6)$

$= -24 + 15 - 12$

$= -21$ //

b) $\vec{a} \cdot (\vec{b} + \vec{c}) \Rightarrow \vec{b} + \vec{c} = \hat{i}(-1+2) + \hat{j}(-4+2) + \hat{k}(2+1)$

$= 1\hat{i} - 2\hat{j} + 3\hat{k}$

* $\vec{a} \cdot (\vec{b} + \vec{c})$

$\Rightarrow 3.(1) + 3(-2) - 2(3)$

$\Rightarrow 3 + (-6) - 6$

$\Rightarrow -9$ //

(c) $\vec{a} \times (\vec{b} + \vec{c})$

$\Rightarrow \vec{b} \times \vec{c} =$

	$\hat{i}$	$\hat{j}$	$\hat{k}$						
b	-1	-4	2	$= +\hat{i}$	-4	2	$-\hat{j}$	-1 2	$+\hat{k}$ -1 -4
c	2	2	1		2	1		2 1	2 2

salah → $= \hat{i}(-4.1 - (2.2)) - \hat{j}(-1.1 - (2.2)) + \hat{k}(-1.2 - (-4.2))$

$= -8\hat{i} + 5\hat{j} + 6\hat{k}$

$\Rightarrow \vec{b} + \vec{c} = \hat{i}(-1+2) + \hat{j}(-4+2) + \hat{k}(2+1)$

$= 1\hat{i} - 2\hat{j} + 3\hat{k}$

$\Rightarrow \vec{a} \times (\vec{b} + \vec{c})$

$\Rightarrow$

	$\hat{i}$	$\hat{j}$	$\hat{k}$						
$\vec{a}$	3	3	-2	$= +\hat{i}$	3 -2	$-\hat{j}$	3 -2	$+\hat{k}$	3 3
$\vec{b}+\vec{c}$	1	-2	3		-2 3		1 3		1 -2

$= +\hat{i}(3.3 - (-2.-2)) - \hat{j}(3.3 - (-2.1)) + \hat{k}(3.-2 - (3.1))$

$= 5\hat{i} - 11\hat{j} - 9\hat{k}$ //

3 Diketahui medan listrik $\vec{E} = 10\hat{i} - 5\hat{j}$ V/m, menembus permukaan $\vec{A} = 2\hat{i} + 2\hat{j}$ m^2, tentukan besar fluxnya $\Phi = \vec{E} \cdot \vec{A}$ dan tentukan sudut antara medan listrik dengan permukaan!

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⇒ $\Phi = \vec{E} \cdot \vec{A}$

$= 10(2) - 5(2)$

$= 20 - 10$

$= 10\ Vm$

$|\vec{E}| = \sqrt{10^2 (-5)^2}$

$= \sqrt{100 + 25}$

$= \sqrt{125}$

$|\vec{A}| = \sqrt{2^2 + 2^2}$

$= \sqrt{4+4}$

$= \sqrt{8}$

⇒ $\cos\theta = \dfrac{\vec{E} \cdot \vec{A}}{|\vec{E}| \cdot |\vec{A}|}$

$= \dfrac{10}{\sqrt{125} \cdot \sqrt{8}}$

$= \dfrac{10}{10\sqrt{10}} \cdot \dfrac{\sqrt{10}}{\sqrt{10}}$

$= \dfrac{10\sqrt{10}}{100}$

$= \dfrac{\sqrt{10}}{10}$ KW I

$\theta = \arccos \dfrac{\sqrt{10}}{10}$ KW I

4. Sebuah muatan positif $q = 2\,\mu C$ bergerak dengan kecepatan $v = 4 \times 10^8$ m/s tegak lurus di daerah medan magnet 0,01 tesla. Hitunglah besar gaya magnet F dan kemana arah gaya magnet F? (Gunakan $\vec{F} = q\vec{v} \times \vec{B}$)

⇒ $\vec{F} = q\vec{v} \times \vec{B}$

⇒ $|\vec{F}| = q|\vec{v}|\,|\vec{B}| \cdot \sin\theta$

$= 2 \times 10^{-6} C \times 4 \times 10^8\ m/s \times 1 \times 10^{-2} \cdot 1$

$= 8\ N$

Arah gaya magnet ke atas

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